Robustness Analysis of Quantum Teleportation under Environmental Noise and Eavesdropping Attacks

Abstract

Quantum teleportation is a foundational protocol for quantum networks, yet its performance is highly dependent on the integrity of the shared entanglement. This research provides a technical evaluation of teleportation robustness by simulating common real-world degradations using the Qiskit framework. We specifically model "partial attacks" (eavesdropping) of varying strengths and hardware-like decoherence to measure their impact on state fidelity and trace distance.

Our simulations demonstrate that entanglement distribution is the most vulnerable phase of the protocol. Quantitative results show that hardware-like noise reduces state fidelity to approximately 0.82, while moderate eavesdropping attempts further degrade the quality of the teleported state to between 0.6 and 0.7. We also examine the correlation between attack strength and fidelity loss, establishing that fidelity monitoring is an effective practical mechanism for detecting unauthorized measurements. This analysis highlights the critical trade-offs between communication security and state quality, providing insights essential for the development of secure and energy-efficient quantum communication infrastructure.